

```

start transaction;
select status from seat where seatID=6 for update;
update seat set status="booked" where seatID=6;
commit;
/*lost update user A*/
start transaction;
update seat set status="booked" where seatID=6;
/*user B*/
start transaction;
update seat set status="reserved" where seatID=6;
commit;
/*user A */
commit;
select *from seat where seatID=6 for update;
/*lost dirtydata user A*/
start transaction;
update seat set status="booked" where seatID=6;
select *from seat;
/*lost dirtydata user B*/
select *from seat where seatID=6;
/*user A*/
rollback;
select *from seat;
set transaction isolation level read committed;
/*lost deadlock user A*/
start transaction;
update seat set status="booked" where seatID=4;
/*user B*/
start transaction;
update seat set status="booked" where seatID=6;
/*user A*/
update seat set status="booked" where seatID=6;
/*user B*/
update seat set status="booked" where seatID=4;
show engine InnoDB status;
/*security and access control*/

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/*Admin*/

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create role Admin;
grant all privileges ON passengers to Admin;
grant all privileges ON flight to Admin;
grant all privileges ON seat to Admin;
grant all privileges ON booking to Admin;
grant all privileges ON payment to Admin;
/*Airline staff*/
create role Airline_staff;
grant select,insert,update on flight to Airline_staff;
grant select,update on seat to Airline_staff;
/*customer*/

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create role customer;

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grant select ON flight to customer;
grant select ON seat to customer;
grant select ON booking to customer;
grant select ON payment to customer;
/*assign roles to users*/
grant Admin to user1;
grant Airline_staff to user2;
grant customer to user3;
/*password security(hashing)*/

```

```

create table user(
userid int primary key,
username varchar(20),
password_hash varchar(200)
);
insert into user(userid,username,password_hash) values(2,"dani",SHA2("mypassword123",256)),
(3,"abula",SHA2("mypassword456",256)), (4,"abrish",SHA2("mypassword701",256));

```

Select *from user;

userid	username	password_hash
2	dani	6e659deaa85842cdabb5c6305fcc40033ba43772ec00d45c2a3c921741a5e377
3	abula	415858d6b1a389558e5fd3b0b661c223bcee5825b6e75e13e2e9de4e5611d0c2
4	abrish	108796d13303971648ac40ebacdfa9658ba5423e70cce000aa04b64c9902e165

```

/*encrypt sensetive data (payment)*/
alter table payment modify encrypted_details varbinary(200);
update payment set encrypted_details=aes_encrypt("1234-8765-8907","key431")where
paymentid=1;
/*audit logging*/

```

```

create table audit_log(
auditid int primary key auto_increment,
action varchar(30),
userid int,
timestamp timestamp default current_timestamp
);

```

select *from audit_log;

auditid	action	userid	timestamp
1	new booking created	2	4/15/2026 13:10
2	booking cancelled	3	4/15/2026 13:10
3	payment completed	4	4/15/2026 13:10

```

/*trigger booking tracking*/
create trigger log_booking after insert on booking for each row insert into audit_log
(action,userid)
values('new booking created', 2),('booking cancelled', 3),('payment completed', 4);

```